All mixed identities are singular in groups with no algebraicity

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joint work with Michael Pinsker

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Outline

1. Background
   - Equations in groups
   - Mixed identities
2. On algebraic closure
3. The result
4. Strategy
5. A topological extra
6. Bibliografy

Words with constants

$G :=$ a group.

Definition (Word with constants)

A **word with constants** for G in r many variables is

$$w(x_1, \ldots, x_r) := \gamma_n \; x_{\iota_n}^{\epsilon_n} \; \gamma_{n-1} \; x_{\iota_{n-1}}^{\epsilon_{n-1}} \; \cdots \; \gamma_1 \; x_{\iota_1}^{\epsilon_1} \; \gamma_0$$

for $\gamma_0, \ldots, \gamma_n \in G$, $\epsilon_1, \ldots, \epsilon_n \in \{-1, 1\}$, and $\iota_1, \ldots, \iota_n \in \{1, \ldots, r\}$.

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I.e., a **term** in $\mathcal{L} := \{\cdot, ^{-1}, (\gamma)_{\gamma \in G}\}$.

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I.e., $w \in G * F_r$.

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w is **singular** if "forgetting constants" and reducing we get 1:

$$y^{-1} x \gamma_2 x^{-1} \gamma_1 y \xRightarrow[\text{constants}]{\text{forget}} y^{-1} x x^{-1} y \mapsto y^{-1} y \mapsto 1 \, .$$

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Otherwise, w is **regular** (a.k.a. non-singular).

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w defines a **word map**: $w : G^r \to G \ (g_1, \ldots, g_r) \mapsto w(g_1, \ldots, g_r)$.

- when can we **solve** w?
  $\exists g_1, \ldots, g_r \in G$ s.t. $w(g_1, \ldots, g_r) = 1$;
- when is w a **mixed identity**?
  $\forall g_1, \ldots, g_r \in G \ w(g_1, \ldots, g_r) = 1$.

These notions are in tension!
$\forall \overline{g} \in G \ w(\overline{g}) = 1 \ \Rightarrow$ for $\gamma \in G \setminus \{1\}$, $\nexists \overline{g} \in G \, \gamma w(\overline{g}) = 1$.

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Conjecture (Klyachko and Thom 2017)

G a group, $w(x_1, \ldots, x_r)$ regular.
Then, there is $H \supseteq G$ s.t. $\exists h_1, \ldots, h_r \in G \; w(h_1, \ldots, h_r) = 1$.

- $r = 1$ this is the Kervaire-Laudenbach Conjecture:
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Theorem (OpenAI 2026 🤖, presumably?)

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Theorem (Adeleke and Holland 1994)

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Theorem (Adeleke and Holland 1994)

$G := \mathrm{Aut}(\mathbb{Q}; <)$. w with no constants and non-trivial.
Then, $w(G^r) = G$.

Theorem (Mycielski 1987; Lyndon 1990)

$G := \mathrm{Sym}(\mathbb{N})$. w with no constants and $w \neq x^n$ $(n \geq 2)$.
Then, $w(G^r) = G$.

An interlude on oligomorphicity

$\mathrm{Aut}(\mathbb{Q};<)$ and $\mathrm{Sym}(\mathbb{N})$ are both oligomorphic:

Definition (Oligomorphic)

$|\Omega| = \aleph_0$. $G \curvearrowright \Omega$ is **oligomorphic** if for all $n \in \mathbb{N}$, $G \curvearrowright \Omega^n$ has finitely many orbits in the action $g \cdot (a_1, \ldots, a_n) \mapsto (ga_1, \ldots, ga_n)$.

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Ryll-Nardzweski: $G = \mathrm{Aut}(M)$ is oligomorphic $\Leftrightarrow$ M is ω-categorical. Many examples are **homogeneous**: every isomorphism between finite substructures of M extends to an automorphism.

Mixed identities

Definition (MIF and lawless)

$w(x_1, \ldots, x_r)$ is **a mixed identity** if $w(G^r) = 1$.
A mixed identity without constants is a **law**.
G is **mixed-identity-free (MIF)**: no (non-trivial) mixed identities.
G is **lawless**: no (non-trivial) laws.

- Mixed identities describe the structure of a group/elements:

$$\forall x\ x^2 = 1 \quad \text{or} \quad \forall x\ xcx^{-1}c^{-1} = 1$$

- we would expect "rich group"$\Rightarrow$ "few" mixed identities.

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$G \curvearrowright \Omega$ *oligomorphic* $\Rightarrow$ G *is lawless.*

Theorem (Melles and Shelah 1994)

M *a saturated structure*[*] $\Rightarrow$ $\mathrm{Aut}(M)$ *is lawless.*

[*] **saturated:** types over subsets $A \subseteq M$ with $|A| < |M|$ are realised in M.

Mixed identities in the oligomorphic/homogeneous context

Several automorphism groups of homogeneous structures are MIF:

- random graph, generic triangle free graph, Urysohn space (Etedadialiabadi, Gao, Le Maître, and Melleray 2021);
- transitive with free amalgamation (Ghadernezhad and de la Nuez González 2019; Bodirsky, Schneider, and Thom 2025);
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Sometimes we have mixed identities!

- $\mathrm{Sym}(\mathbb{N})$: $[x,t]^6$ where t is a transposition
  (Hull and Osin 2016);
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In the first two cases, we know all mixed identities are singular!

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  (Bodirsky, Schneider, and Thom 2025);
- $\mathrm{GL}(\aleph_0, \mathbb{F}_q)$: all mixed identities are singular
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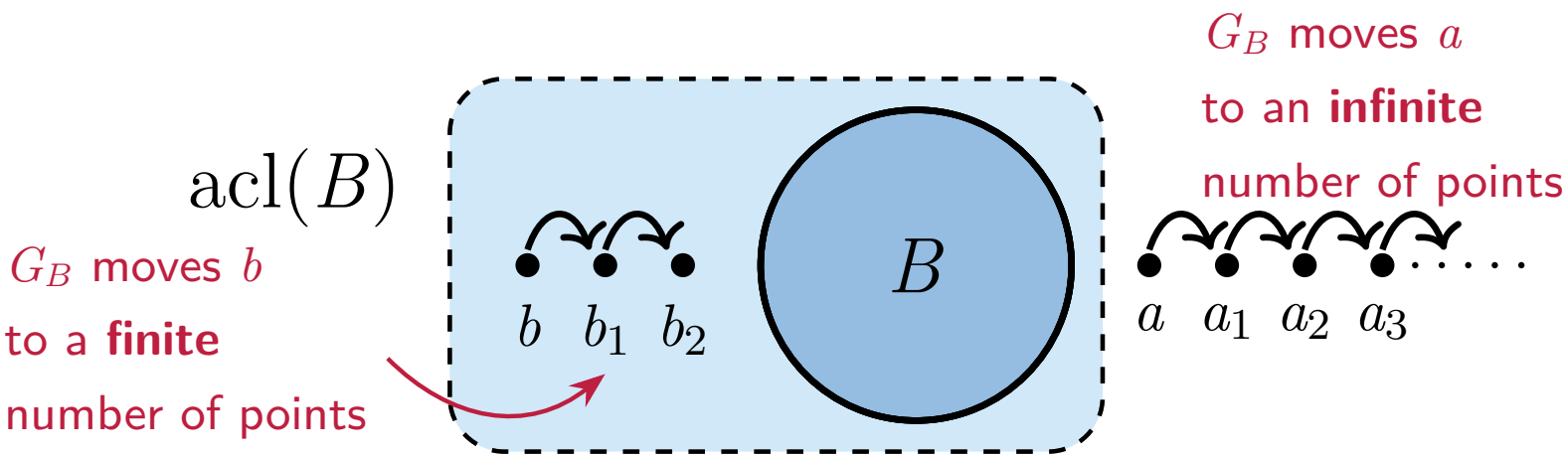
We want general conditions that imply all mixed identities are singular.

Algebraic closure

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Lemma (Π. M. Neumann 1976)

$G \curvearrowright \Omega$. $A, B \subseteq \Omega$ finite and $A \cap \mathrm{acl}(\emptyset) = \emptyset$.
Then, there is $g \in G$ such that $(g \cdot A) \cap B = \emptyset$.

Algebraic closure

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Theorem (Abért 2005)

$G \curvearrowright \Omega$ with no algebraicity. Then, G is lawless.

Our result

Theorem (MP 2026)

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- automorphism groups of homogeneous structures whose age has the **strong amalgamation property**:
  - recovers most previous results;
  - confirms the Bodirsky, Schneider, and Thom 2025 conjecture for several cases beyond previous techniques:
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- Weakly branch groups (include Grigorchuk's group).
  Has mixed identities (Jacobson 2021; Le Boudec and Matte Bon 2022);
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$$\{(g_1, \dots, g_r) \in G^r \mid w(g_1, \dots, g_r) \neq 1\}$$
is dense in G^r wrt the topology of pointwise convergence.[†]
- We do not need the action to be faithful.

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Proof strategy[‡]

For $g \in G$, $a_0 \in \Omega$, the images of a_0 with respect to subwords of w create a sequence $(a_0, a_1, a_1', \ldots, a_n, a_n', a_{n+1})$ with

$$g \cdot (a_1, \ldots, a_n) = (a_1', \ldots, a_n') \ .$$

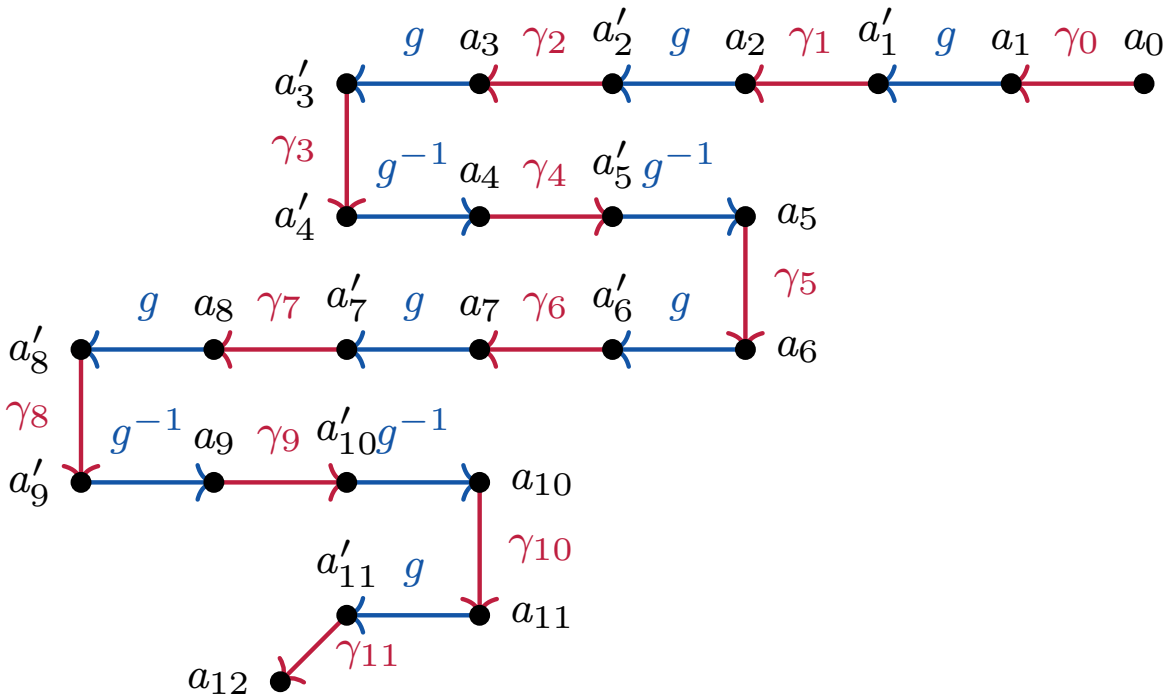
[‡] w with one variable.

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$$w(x) = \gamma_{11} x \gamma_{10} x^{-1} \gamma_9 x^{-1} \gamma_8 x \gamma_7 x \gamma_6 x \gamma_5 x^{-1} \gamma_4 x^{-1} \gamma_3 x \gamma_2 x \gamma_1 x \gamma_0$$

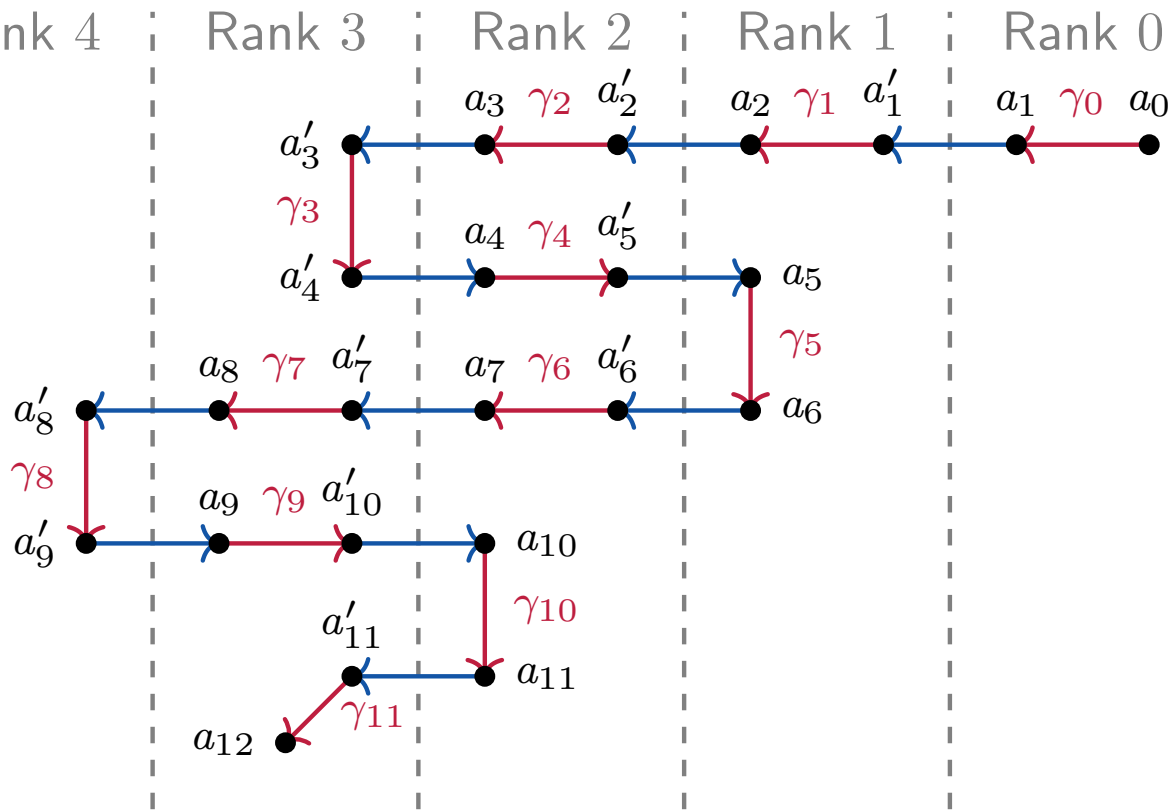


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Rank elements according to their position in $\mathbb{Z}$.

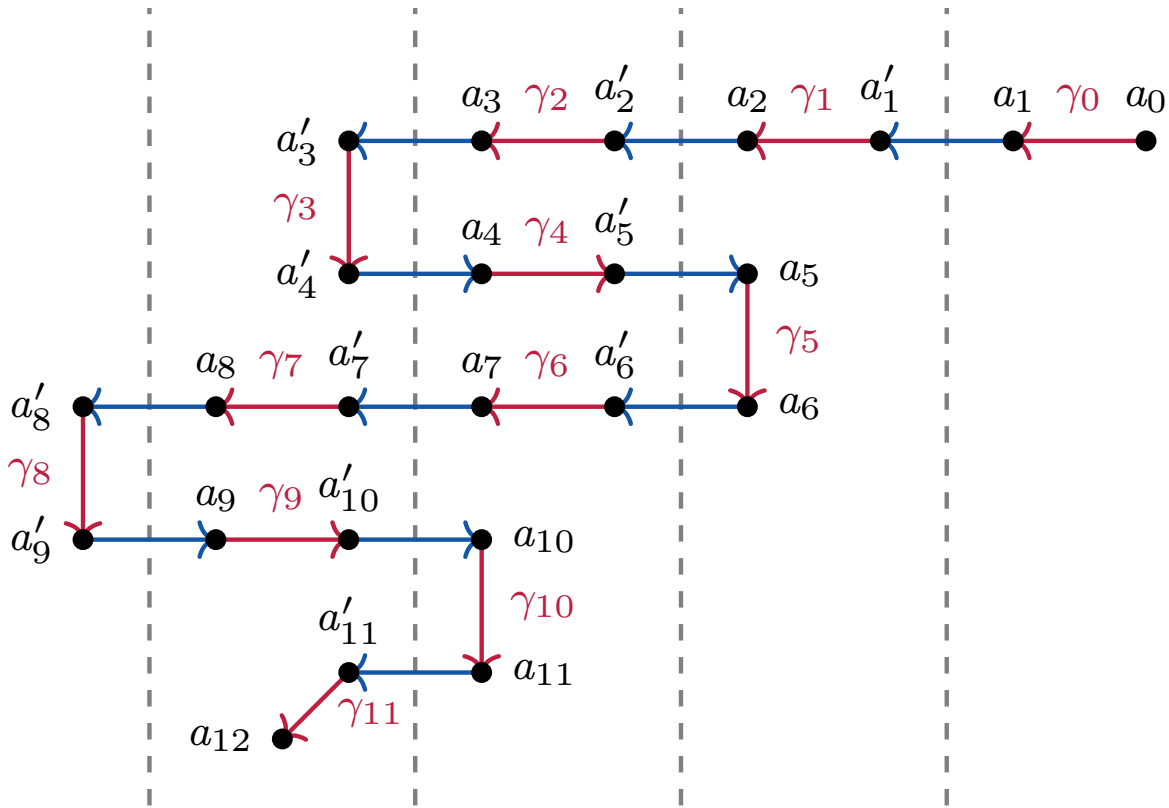


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We want g so that elements of different ranks are independent!

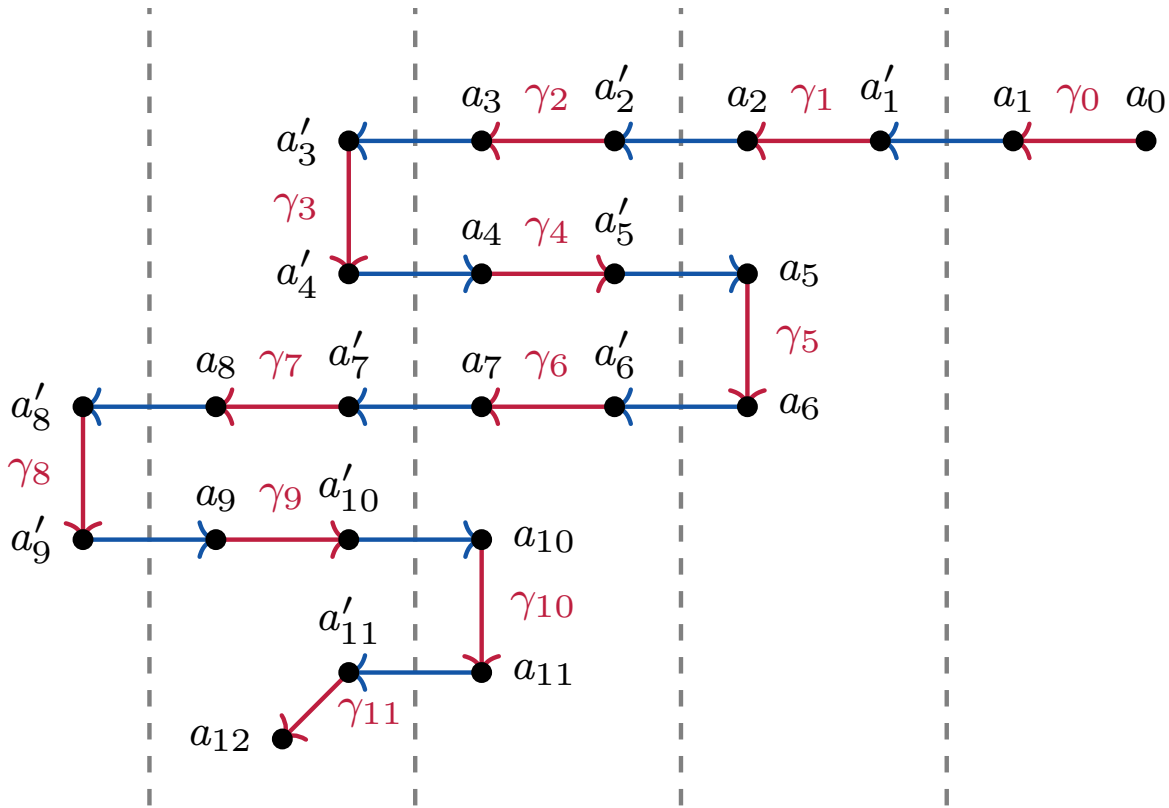


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We cannot guarantee elements of same rank to be distinct!

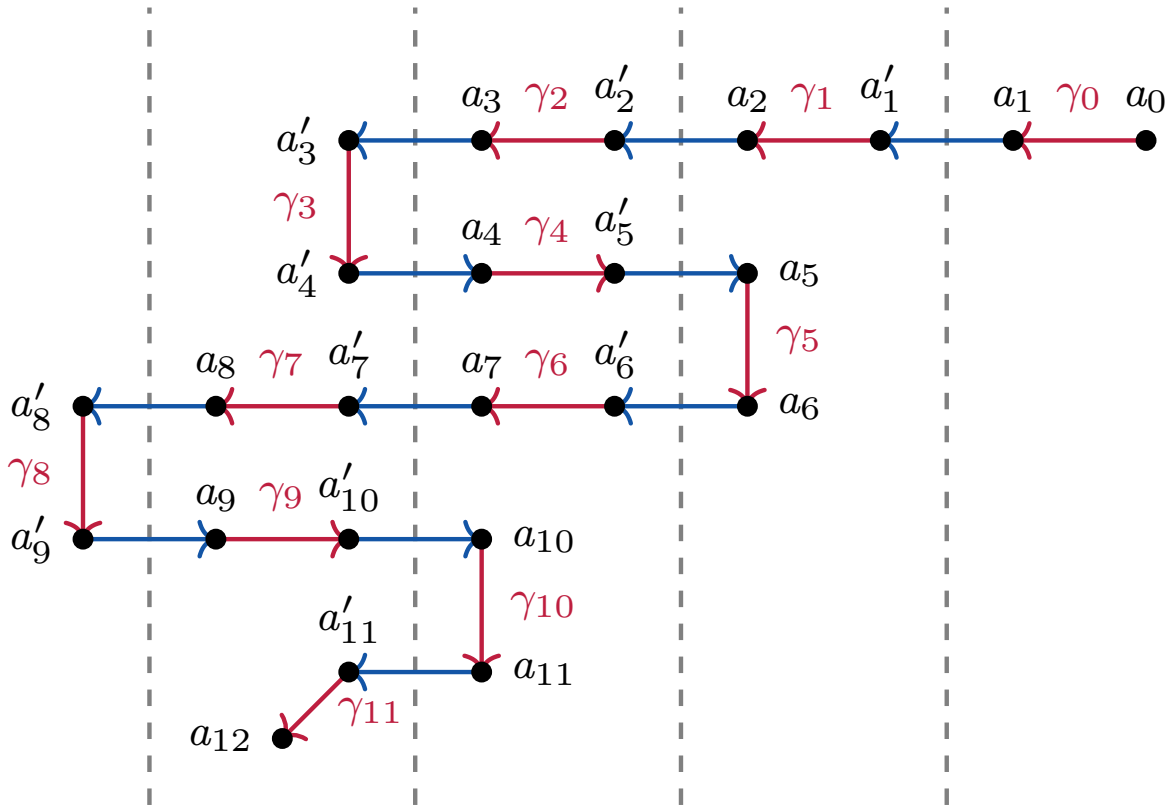


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Use Neumann's Lemma to guarantee independence (looking ahead!)

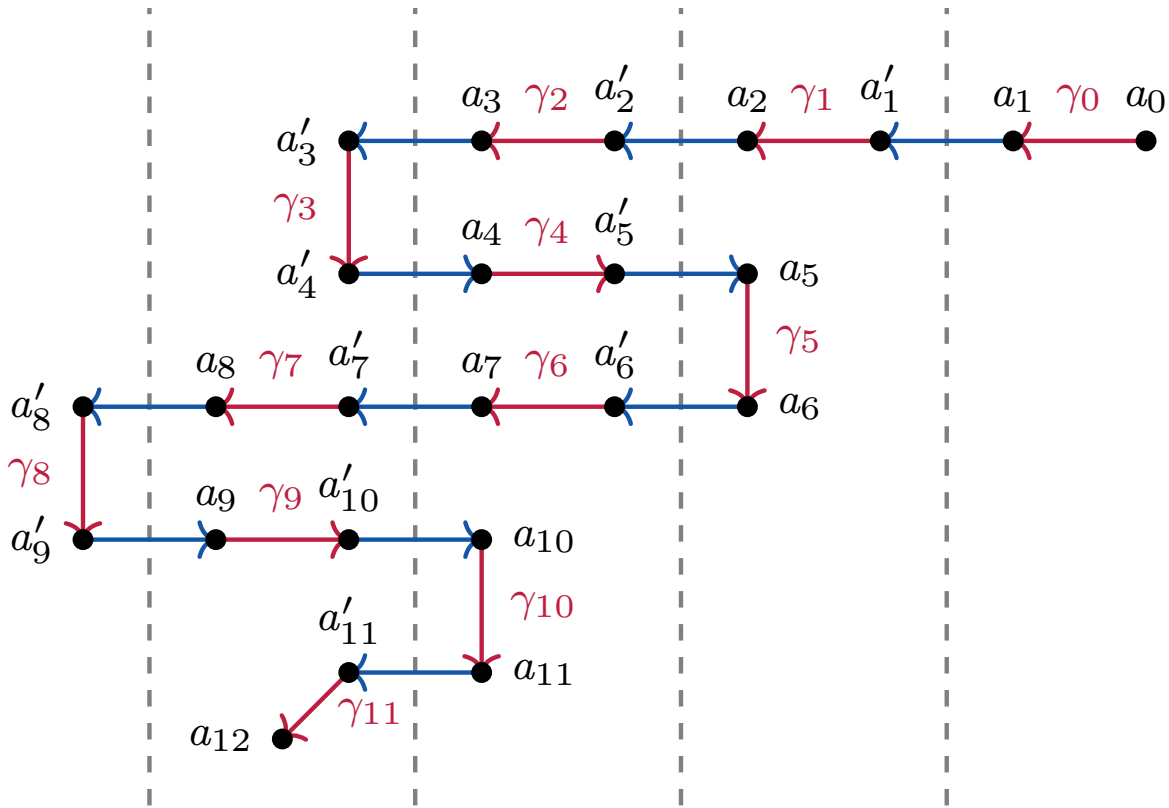


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Induction proof: challenge is balancing inductive hypothesis!



Geometric properties underlying the proof

With algebraicity, we need further geometric properties and a strengthening of Neumann's Lemma.

Definition (pregeometry)

(Ω, cl) for $\mathrm{cl} : \mathcal{P}(\Omega) \to \mathcal{P}(\Omega)$ is a **pregeometry** if it satisfies:

(REFLEXIVITY)	$A \subseteq \mathrm{cl}(A)$;
(TRANSITIVITY)	$\mathrm{cl}(\mathrm{cl}(A)) = \mathrm{cl}(A)$;
(MONOTONICITY)	if $A \subseteq B$, then $\mathrm{cl}(A) \subseteq \mathrm{cl}(B)$;
(FINITE CHARACTER)	$\mathrm{cl}(A) = \bigcup\{\mathrm{cl}(C) \mid C \subseteq A \text{ finite}\}$
(EXCHANGE)	if $a \in \mathrm{cl}(A \cup \{b\}) \setminus \mathrm{cl}(A)$, then $b \in \mathrm{cl}(A \cup \{a\})$.

It is **G-invariant** if for $g \in G$, $\mathrm{cl}(gA) = g\,\mathrm{cl}(A)$.

(Ω, cl) is **modular** if for all closed $A, B \subseteq \Omega$,

$$\dim(A \cup B) = \dim(A) + \dim(B) - \dim(A \cap B)\ .$$

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Higher Neumann pregeometries

Definition

Let $G \curvearrowright \Omega$. G-invariant (Ω, cl) is **higher Neumann** if for each $B, C \subseteq \Omega$ finite, $a \in \Omega$ such that $\mathrm{cl}(a) \cap \mathrm{cl}(B) = \mathrm{cl}(\emptyset)$, and $\Sigma \subseteq G$ finite, there is $g \in G_B$ such that

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Let $G \curvearrowright \Omega$. Let (Ω, cl) be a G-invariant infinite-dimensional higher Neumann modular pregeometry.
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Let $G \curvearrowright \Omega$. Let (Ω, cl) be a G-invariant infinite-dimensional higher Neumann modular pregeometry.
Then all mixed identities of G are singular.

Applies to $\mathrm{GL}(\kappa, K)$ and $\mathrm{PGL}(\kappa, K)$ for $\kappa \geq \aleph_0$, recovering results of Bradford, Schneider, and Thom 2023.

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Bradford, Ersoy, Schneider, and Thom 2026:
also yields the result for $\mathrm{Sp}(\aleph_0, \mathbb{F}_q)$, $\mathrm{SU}(\aleph_0, \mathbb{F}_q)$, $\mathrm{O}(\aleph_0, \mathbb{F}_q)$.

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▸ More on the proof

Microsupported actions

Several of the examples coming from topology are microsupported:

Definition

$\mathbf{\Omega} = (\Omega; \tau)$ Hausdorff, $G \leq \mathrm{Homeo}(\mathbf{\Omega})$. G is **microsupported** if for every non-empty open set $U \subseteq \Omega$, $G_{\Omega \setminus U} \neq \{1\}$.

Theorem (Nekrashevych 2022)

If G is microsupported, then G is lawless.

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Different proof idea: lifting lawlessness through microsupport.

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Some questions

Questions you may enjoy (please do not feed to 🤖):

Question

Let M be saturated. Are all mixed identities of $\mathrm{Aut}(M)$ singular?

Question

Is there a lawless group with a non-singular mixed identity?

Thank you for your time!

Recap:

- All mixed identities being singular is a strong form of lawlessness;
- We prove this for:
  - groups with no algebraicity;
  - groups with microsupported actions.
- Key new idea: use global geometric properties of actions instead of properties of constants (controlling products of constants).

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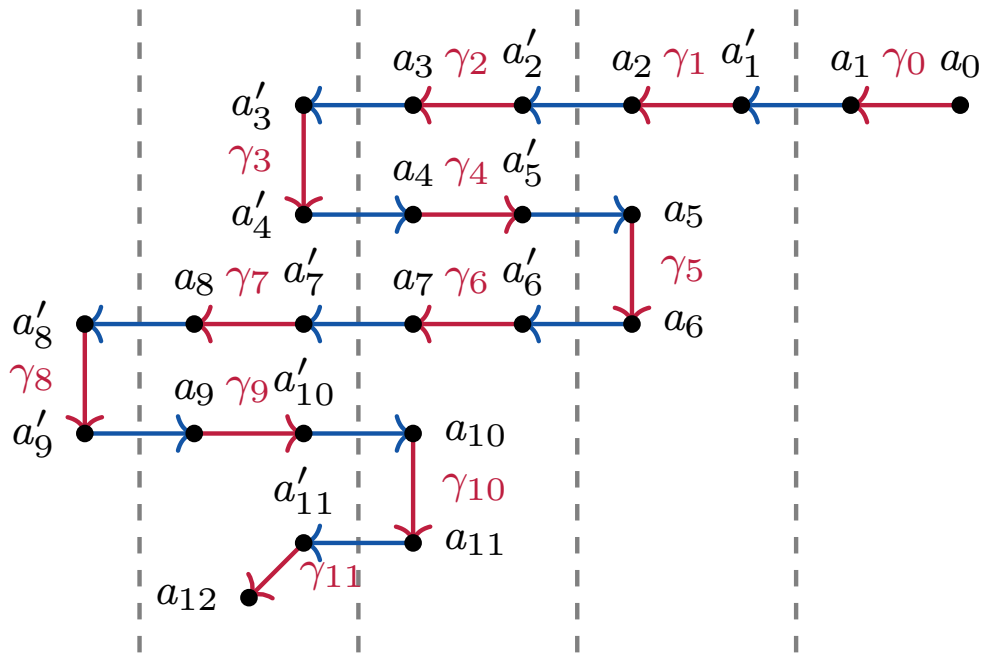
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Why these geometric conditions?

Induction splits into two cases:

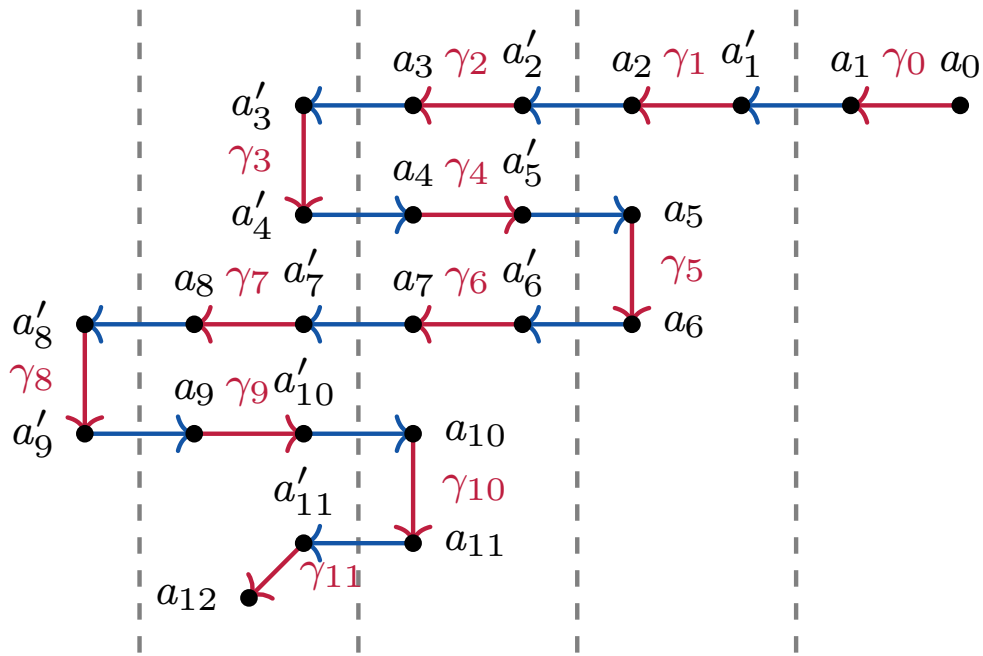
- a_{n+1} is in the closure of previous elements;
- a_{n+1} is not in the closure of previous elements.



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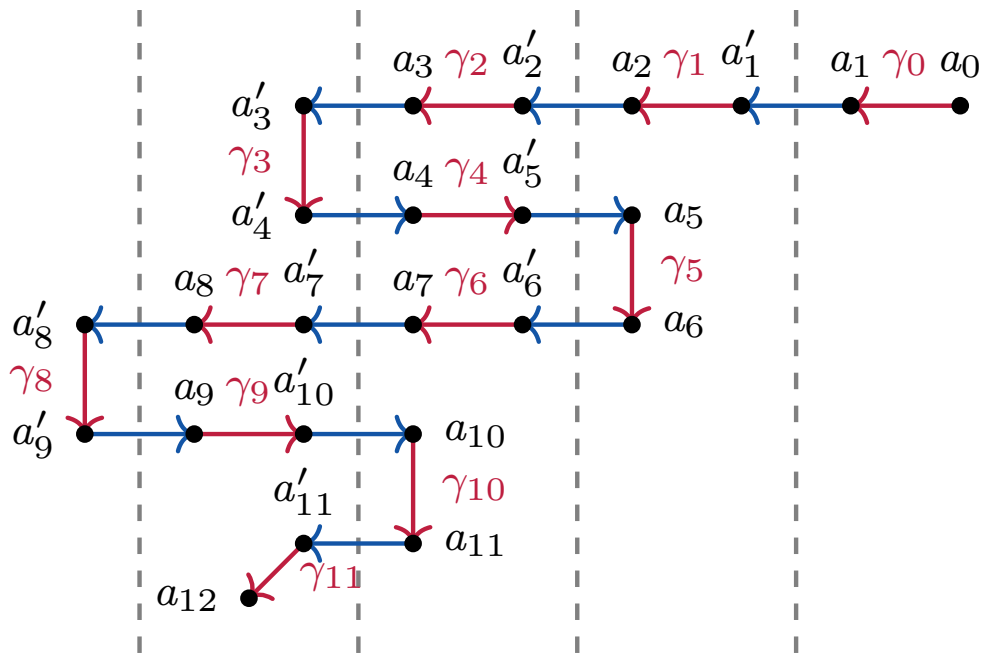
- a_{n+1} is in the closure of previous elements
  $\Rightarrow$ By modularity+IH,
  a_{n+1} is in closure of previous elements of its rank;
  we already know where it has to move
  (with IH guaranteeing independence of columns).
- a_{n+1} is not in the closure of previous elements.



Why these geometric conditions?

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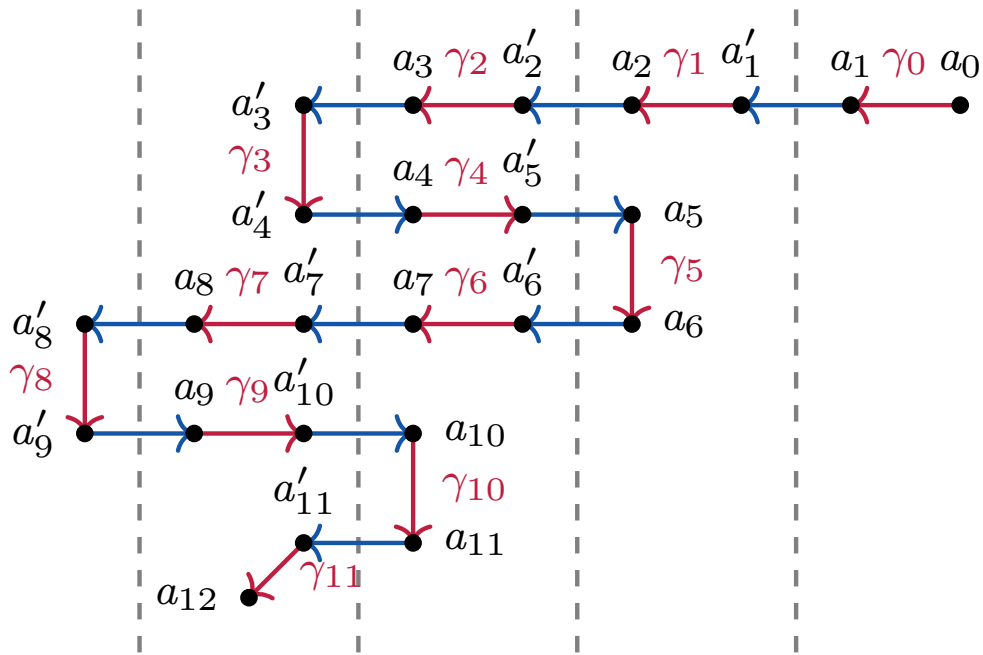
- a_{n+1} is in the closure of previous elements;
- a_{n+1} is not in the closure of previous elements
  $\Rightarrow$ By higher Neumann+pregeometry, move a_{n+1} and all of its translates to guarantee independence of columns.



Why these geometric conditions?

Induction splits into two cases:

- a_{n+1} is in the closure of previous elements;
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▸ Go back

The microsupported proof

Proof.

Let $w := \gamma_n \; x_{\iota_n}^{\epsilon_n} \; \gamma_{n-1} \; x_{\iota_{n-1}}^{\epsilon_{n-1}} \; \dots \; \gamma_1 \; x_{\iota_1}^{\epsilon_1} \; \gamma_0$.

$u := w$ after forgetting constants.

Let $\xi_k := \gamma_k \gamma_{k-1} \cdots \gamma_0$; $\Sigma := \{1, \xi_0, \dots, \xi_n\}$.

By microsupport: there is $V \subseteq \Omega$ open such that any two members of Σ either agree on V or have disjoint images.

$G_{\Omega \setminus V} \curvearrowright V$ is also microsupported. By lawlessness, there are $h_1, \dots, h_r \in G_{\Omega \setminus V}$ such that $u(h_1, \dots, h_r) \neq 1$.

Set $p_i := \prod_{j \in D} \xi_j h_i \xi_j^{-1}$ (D selects one index per disjoint image).

- For $a \in V$, $w(p_1, \dots, p_r)(a) = \xi_n u(h_1, \dots, h_r)(a)$.

By construction of V (and $1, \xi_n \in \Sigma$):

- (EITHER) $\xi_n(V) \cap V = \emptyset$: $w(p_1, \dots, p_n)(V) \cap V = \emptyset$;
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Back to presentation

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